

1. Introduction

1.1 Program Overview

The program CUSTPROC is a mainframe COBOL batch/online program designed to perform core customer business processing. It handles data ingestion, variable preparation, validation, and communicates with various backend services. It was authored by ANTIGRAVITY.

1.2 Objectives

The primary objectives of CUSTPROC are: - Retrieve customer profile information from DB2 databases and IMS databases. - Validate transactions and perform formatting on sensitive customer fields. - Log status messages and transfer buffers to external message queues via IBM MQ. - Present processed details on user screens via CICS interfaces.

1.3 Scope

This program serves as a processing module within the Customer Management subsystem. It manages VSAM database files, processes inbound linkage section parameters, and performs data staging.

1.4 Assumptions and Constraints

- **Assumptions:** DB2 tables and IMS databases are online and accessible. CICS transaction manager and MQ series queues are operational.
- **Constraints:** Fixed-length processing windows, strictly typed variable schemas defined in copybooks, and compliance with mainframe memory layouts.

2. Database Details

2.1 DB2 Tables

The program queries or updates the following database tables: - DB2_CUST_TABLE

2.2 IMS Segments

The program interfaces with IMS databases using the following segment search arguments (SSAs) or IO areas: - Segment/IO Area: CUST-SEG-IO-AREA - Segment/IO Area: CUST-SEG-SSA

2.3 IDMS Records

The program retrieves or modifies the following IDMS database records: - Record:
CUST-RECORD-IDMS

2.4 External Module and Copybook Dependencies

The program relies on the following external files, copybooks, subprograms, and mapsets:

Copybooks (Data Structures)

- `CUSTOMER-RECORD`: Imported layout definition defining record formats.
- `DB-STATUS`: Imported layout definition defining record formats.

No external program calls detected in procedure division.

VSAM and File Resources

- `CUSTFILE` (Dataset/DD: `VSAM-CUST-FILE`): Logical file reference for storage/transfer operations.

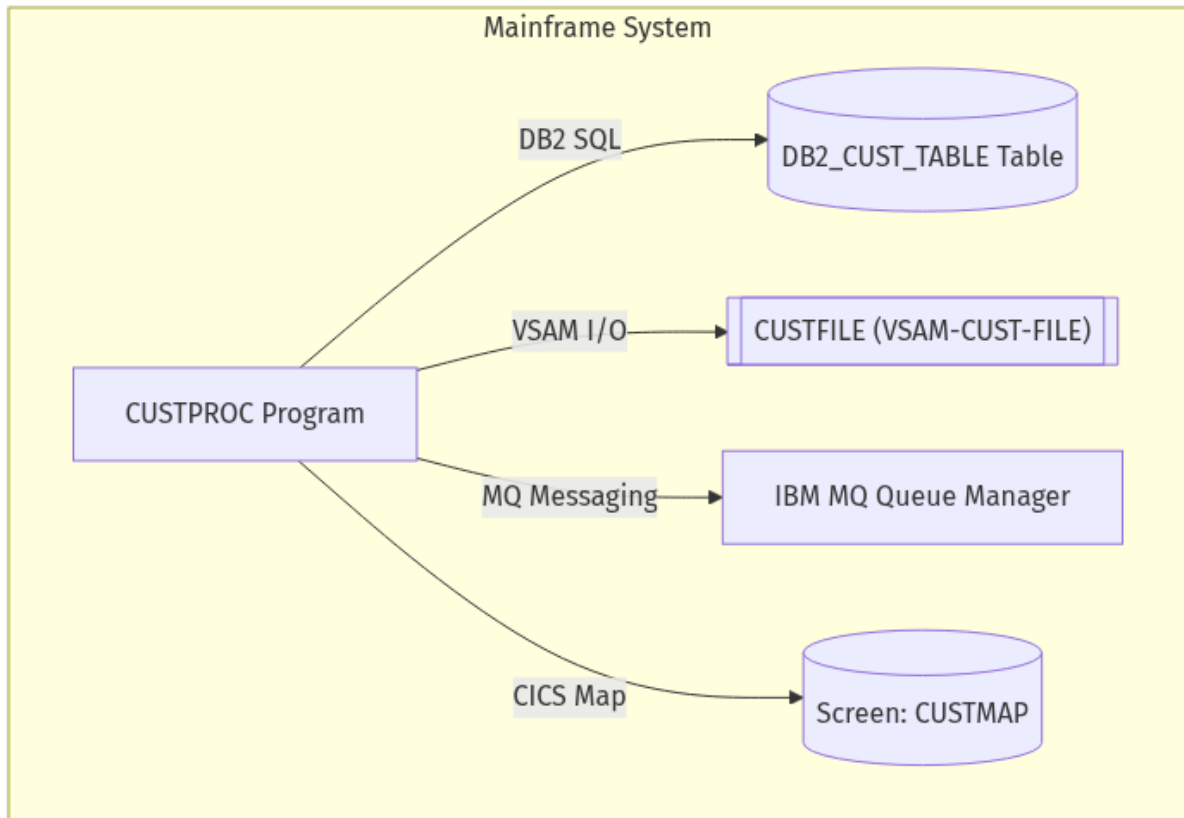
CICS Screens and Maps

- Map `CUSTMAP`: User interface screen layout.
- Mapset `CUSTSET`: Set of screen map configurations.

3. System Architecture

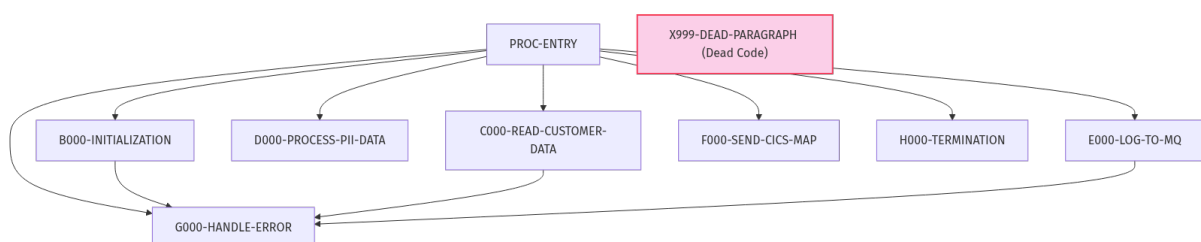
3.1 Component Diagram

The diagram below outlines the external interfaces and components interfacing with the CUSTPROC program:



3.2 Control Flow Diagram

The diagram below shows the control flow and call relationships between the internal paragraphs of the program:



4. Detailed Design

4.1 Program Structure

4.1.1 Paragraph and Section Architecture

The following table details all sections and paragraphs defined in the program:

Paragraph / Section	Start Line	End Line	Section Group	Description
PROC-ENTRY	42	53	A000-MAIN-PROCESSING	Initial processing logic
B000-INITIALIZATION	54	63	A000-MAIN-PROCESSING	Initial processing logic
C000-READ-CUSTOMER DATA	64	89	A000-MAIN-PROCESSING	Queries DB2, Calls IMS, Accesses VSAM files
D000-PROCESS-PII-DATA	90	95	A000-MAIN-PROCESSING	Initial processing logic
E000-LOG-TO-MQ	96	107	A000-MAIN-PROCESSING	Sends outputs to IBM MQ
F000-SEND-CICS-MAPS	108	116	A000-MAIN-PROCESSING	Sends CICS Maps
G000-HANDLE-ERROR	117	121	A000-MAIN-PROCESSING	Initial processing logic
H000-TERMINATION	122	124	A000-MAIN-PROCESSING	Initial processing logic
X999-DEAD-PARAGRAPH	125	128	A000-MAIN-PROCESSING	Initial processing logic

4.1.2 Data Division Layout and Key Variable Structures

WORKING-STORAGE SECTION Variables

Variable Name	Level	PIC Clause	Initial Value	Source Copybook
CUSTOMER-RECORD	1	N/A	N/A	CUSTOMER-RECORD
CUST-ID	5	X(10)	N/A	CUSTOMER-RECORD
CUST-NAME	5	X(30)	N/A	CUSTOMER-RECORD
CUST-SSN	5	X(9)	N/A	CUSTOMER-RECORD
CUST-CONTACT	5	N/A	N/A	CUSTOMER-RECORD
CUST-PHONE	10	X(10)	N/A	CUSTOMER-RECORD
CUST-EMAIL	10	X(40)	N/A	CUSTOMER-RECORD
CUST-ADDRESS	5	N/A	N/A	CUSTOMER-RECORD
CUST-STREET	10	X(30)	N/A	CUSTOMER-RECORD
CUST-CITY	10	X(20)	N/A	CUSTOMER-RECORD
CUST-STATE	10	X(2)	N/A	CUSTOMER-RECORD
CUST-ZIP	10	X(5)	N/A	CUSTOMER-RECORD
CUST-ACCOUNT-NUM	5	X(12)	N/A	CUSTOMER-RECORD
CUST-BALANCE	5	S9(7)V99	N/A	CUSTOMER-RECORD
DB-STATUS	1	N/A	N/A	DB-STATUS

SQL - CODE - VAL	5	S9(9)	N/A	DB-STATUS
IMS-STATUS-CODE	5	X(2)	N/A	DB-STATUS
IDMS-ERR-CODE	5	X(4)	N/A	DB-STATUS
MQ-REASON-CODE	5	S9(9)	N/A	DB-STATUS
VSAM-FEEDBACK	5	X(2)	N/A	DB-STATUS
ERR-MSG	5	X(80)	N/A	DB-STATUS
WS-VARIABLES	1	N/A	N/A	Local
VSAM-STATUS	5	X(2)	N/A	Local
WS-PII-LOG-SSN	5	X(9)	N/A	Local
WS-MQ-MSG-BUFF	5	X(200)	N/A	Local
WS-SYS-DATE	5	X(8)	N/A	Local
IMS-CONSTANTS	1	N/A	N/A	Local
IMS-GU	5	X(4)	GU	Local
IMS-GN	5	X(4)	GN	Local

LINKAGE SECTION Variables (Input/Output Parameters)

Variable Name	Level	PIC Clause	Source Copybook
LK-INPUT-DATA	1	X(50)	Local

4.2 Algorithms

4.2.1 Overall Program Logic

The CUSTPROC program logic coordinates execution through distinct structural stages: 1.

Initialization: The program prepares internal processing variables, initializes database cursor areas, and opens any referenced files or screen resources. It establishes base variables needed for loop processing. 2. **Main Processing Loop:** It reads input data sequentially (either via files or cursor fetches). For each item: it performs validations, retrieves related entity details from DB2 tables/IMS databases, propagates key data elements, and handles business updates or queuing operations. 3. **Termination and Shutdown:** Once input limits or end-of-file indicators are hit, the program cleanly terminates. It closes all active database cursors, releases VSAM files, executes final output prints/displays, and runs GOBACK/STOP RUN.

4.2.2 Key Algorithmic Details

- **Branching & Decisions:** The control path is heavily governed by conditional branching checking SQL execution feedback (SQLCODE) or VSAM file statuses. For example, successful reads proceed to data manipulation while failures trigger specific error recovery paragraphs.
- **Data Translation & Formatting:** Variables parsed from database sources are mapped to intermediate fields, formatted according to PIC specifications, and stored in target records for output (e.g. IBM MQ buffers or CICS map displays).
- **Fault Tolerance:** The logic implements dedicated paragraph paths for error capture, logging, and application recovery to prevent uncontrolled program abends.

4.3 Input/Output Specifications

The primary files and UI interfaces are defined below:

Interface Channel	Technical Resource / DD Name	Description	Data Format	Direction
VSAM File	VSAM-CUST-FILE	File variable <u>CUSTFILE</u>	Index Keyed Record	In/Out
IBM MQ	<u>MQ Series</u> Queue	Buffer message transmission	MQ Descriptor Block	Output
CICS Terminal	<u>CUSTMAP</u> Screen	Terminal user-interface map	Screen Layout Fields	Output

4.4 DB2 Database Details

- **Paragraph C000-READ-CUSTOMER-DATA:** Executes `SELECT` on table `DB2_CUST_TABLE`. SQL Query: `sql SELECT CUST_NAME, CUST_SSN, CUST_ACCOUNT_NUM INTO :CUST-NAME, :CUST-SSN, :CUST-ACCOUNT-NUM FROM DB2_CUST_TABLE WHERE CUST_ID = :CUST-ID`

4.5 IMS Database Details

- **Paragraph C000-READ-CUSTOMER-DATA:** Invokes IMS command `GU` using parameters: `CUST-SEG-IO-AREA`, `CUST-SEG-SSA`

4.6 IDMS Database Details

- **Paragraph C000-READ-CUSTOMER-DATA:** Executes IDMS `OBTAIN` on record `CUST-RECORD-IDMS` within Area `CUST-AREA-IDMS`

4.7 Called Sub-routine/Program Details/Copybook

Included Copybooks: - `CUSTOMER-RECORD`: Pulled into `DATA DIVISION` structures. - `DB-STATUS`: Pulled into `DATA DIVISION` structures.

4.8 VSAM File Details

- **Paragraph C000-READ-CUSTOMER-DATA:** Performs `READ` on file variable `CUSTFILE` (Dataset/DD: `VSAM-CUST-FILE`)

4.9 IBM MQ Details

- **Paragraph E000-LOG-TO-MQ:** Executes IBM MQ function `MQPUT` with argument trace: `MQ-HCONN`, `MQ-HOBJ`, `MQ-MD`, `MQ-PMO`, `WS-MQ-MSG-BUFF`, `MQ-REASON-CODE`

4.10 CICS Details

- **Paragraph F000-SEND-CICS-MAP:** Runs `SEND MAP( 'CUSTMAP' )` on map `CUSTMAP` (Mapset: `CUSTSET`). CICS details: `SEND MAP( 'CUSTMAP' ) MAPSET( 'CUSTSET' )`
`FROM( CUSTOMER-RECORD ) ERASE`

4.11 Error Handling

- **Error Paragraph G000 - HANDLE - ERROR:** Handles faults by logging or displaying errors. Code snippet: ``cobol DISPLAY 'ERROR OCCURRED. SQL-CODE=' SQL-CODE-VAL DISPLAY 'VSAM-STATUS=' VSAM-STATUS MOVE 'AN ERROR WAS ENCOUNTERED IN PROCESSING' TO ERR-MSG.

``

5. Interface Design

5.1 External Interfaces

The program exposes external interfaces through data transfer protocols and database links: - **DB2 Database Link**: Communicates using embedded SQL over the mainframe SQL engine. - **VSAM File Access**: Interfaces with datasets on disk through the operating system's Virtual Storage Access Method. - **IBM MQ Queue Manager**: Exchanges messages asynchronously with downstream applications.

5.2 User Interface

The user interface is a CICS character-mode screen designed using CUSTMAP. It displays formatted customer fields, allows basic input parameters, and outputs system alerts on completion.

6. Testing Strategy

6.1 Test Plan

To verify correct behavior, the test plan includes: 1. **Unit Testing**: Stub out external database queries. Verify that paragraphs are performed in the correct sequence. 2. **DB2 Integration Testing**: Verify database query syntax and error codes (such as SQLCODE = -811, 0, or 100). 3. **VSAM File Verification**: Validate VSAM file read operations and handling of 'record not found' exceptions. 4. **CICS Screen Verification**: Test terminal displays through CICS transaction simulators.

6.2 Testing Environment

The test environment requires standard IBM z/OS development environments. This includes: - IBM Developer for z/OS (IDz) or similar IDE. - IBM MQ simulator or test queue managers. - CICS testing region equipped with the compiled map set (CUSTSET). - DB2 test tables mirroring production schemas with masked/synthesized test data.

7. Performance Considerations

7.1 Performance Analysis

The program's performance is dominated by external input/output operations: - **DB2 Access:** SQL query execution uses CPU MIPS. Lack of indexes can trigger table scans. - **IMS & VSAM Reads:** Accessing files/segments causes disk I/O wait states. - **MQ Messaging:** Network/queue synchronization calls introduce thread-blocking latencies.

7.2 Optimization Recommendations

- Ensure the `DB2_CUST_TABLE` has a primary index on `CUST_ID` to avoid costly full table scans.
- Optimize VSAM buffer sizes (`BUFND/BUFNI`) in JCL definitions to cache file index nodes.
- Replace repetitive queries with bulk SELECT cursors where feasible.
- Ensure MQ calls are performed asynchronously (`MQPMO-NO-SYNCPPOINT`) if transactional guarantees are not required.

8. Dead Code Analysis

The control flow reachability analysis has identified **1** unreachable paragraphs:

- `X999 - DEAD - PARAGRAPH`

Risk/Impact: These paragraphs represent dead code. Having dead code increases maintenance overhead, confuses developer comprehension, and can indicate orphaned logical blocks or legacy routines that are no longer in use. It is recommended to safely comment out or delete these blocks after verifying no dynamic/computed performance routes (e.g. dynamic CALLs) target them.

9. Security Risk Assessment

9.1 PII (Personally Identifiable Information) Analysis

9.1.1 PII Data Elements Identification

The following variables contain or handle PII based on name pattern rules:

Variable Name	Level	PIC Clause	PII Classification Reason
CUST-NAME	5	X(30)	Declared definition (from copybook CUSTOMER-RECORD)
CUST-EMAIL	10	X(40)	Declared definition (from copybook CUSTOMER-RECORD)
CUST-ADDRESS	5	N/A	Declared definition (from copybook CUSTOMER-RECORD)
CUST-ACCOUNT-NUM	5	X(12)	Declared definition (from copybook CUSTOMER-RECORD)
CUST-PHONE	10	X(10)	Declared definition (from copybook CUSTOMER-RECORD)
WS-PII-LOG-SSN	5	X(9)	Declared definition
WS-MQ-MSG-BUFF	5	X(200)	Tainted in D000-PROCESS-PII-DATA from CUST-ACCOUNT-NUM

CUST-SSN	5	X(9)	Declared definition (from copybook CUSTOMER-RECORD)
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9.1.2 PII Data Flow Analysis

Static taint tracking shows the propagation of PII data across variable reassignments:

Step	Source Variable	Destination Variable	Code Location (Paragraph)	Instruction Statement
1	CUST-ACCOUNT-NUM	WS-MQ-MSG-BUFF	D000-PROCESS-PII-MOVE	CUST-ACCOUNT-NUM TO WS-MQ-MSG-BUFF

9.1.3 PII Security Analysis

The analysis revealed critical exposures regarding Personally Identifiable Information (PII): -

Unencrypted Transmission: PII data elements like CUST-ACCOUNT-NUM are copied to the MQ buffer and sent over queues without field-level encryption. - **Log Exposure Risk:** Staging PII records in general logging blocks (e.g. WS-PII-LOG-SSN or displays) can leak sensitive data into system spools (SDSF) where unauthorized users can view them. - **Mitigation Recommendations:** Use IBM InfoSphere Guardium or DFSMS/RACF file encryption. Mask fields (e.g. XXX-XX-1234) before writing to message buffers or system logs.

9.1.4 PII Inventory Summary

A summary of PII egress points and risk classifications:

Egress Channel	Data Element	Risk Level	Mitigation Strategy
DB2 SQL (Table: DB2_CUST_TABLE (SELECT))	CUST-NAME	MEDIUM	Apply field masking
DB2 SQL (Table: DB2_CUST_TABLE (SELECT))	CUST-ACCOUNT-NUM	MEDIUM	Apply field masking
DB2 SQL (Table: DB2_CUST_TABLE (SELECT))	CUST-SSN	MEDIUM	Apply field masking

IBM MQ (MQ queue via MQPUT)	WS-MQ-MSG-BUFF	HIGH	Encrypt before transmission
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10. Appendices

10.1 Glossary

Term	Definition
Copybook	A reusable source file that contains data structures or declarations in COBOL.
DB2	IBM's relational database management system for z/OS mainframe environments.
IMS	Information Management System, a database and transaction management system.
VSAM	Virtual Storage Access Method, a high-performance file management system on IBM mainframes.
IBM MQ	Message queue middleware facilitating secure, asynchronous system-to-system communications.
CICS	Customer Information Control System, an online transaction manager for mainframe systems.
PII	Personally Identifiable Information; sensitive data elements that can identify individuals.
Dead Code	Program blocks or paragraphs that cannot be reached or executed under any conditions.

10.2 Data Structures

Copybook CUSTOMER-RECORD Layout

```
01 CUSTOMER-RECORD. 05 CUST-ID PIC X(10). 05 CUST-NAME PIC X(30). 05 CUST-SSN  
PIC X(9). 05 CUST-CONTACT. 10 CUST-PHONE PIC X(10). 10 CUST-EMAIL PIC X(40). 05  
CUST-ADDRESS. 10 CUST-STREET PIC X(30). 10 CUST-CITY PIC X(20). 10 CUST-STATE  
PIC X(2). 10 CUST-ZIP PIC X(5). 05 CUST-ACCOUNT-NUM PIC X(12). 05 CUST-BALANCE
```

```
PIC S9(7)V99.
```

Copybook **DB-STATUS** Layout

```
01 DB-STATUS. 05 SQL-CODE-VAL PIC S9(9). 05 IMS-STATUS-CODE PIC X(2). 05  
IDMS-ERR-CODE PIC X(4). 05 MQ-REASON-CODE PIC S9(9). 05 VSAM-FEEDBACK PIC X(2).  
05 ERR-MSG PIC X(80).
```

10.3 References

- IBM Enterprise COBOL for z/OS Language Reference
- DB2 SQL Reference Guide
- CICS Application Programming Guide
- IBM MQ Application Programming Reference